JUNIOR SCHOOL LEARNING HUB

GRADE 9 INTEGRATED SCIENCE

DETAILED FULL-YEAR LEARNER NOTES

Terms 1, 2 and 3

Prepared for Junior School Learning Hub
Original teacher-prepared learning resource | CBC-aligned study support

Learner's Name: _____
Grade/Stream: _____
School: _____

Curriculum coverage: Structure of the Atom; Metals and Alloys; Water Hardness; Nutrition in Plants; Nutrition in Animals; Reproduction in Plants; Interdependence of Life; Curved Mirrors; Waves.

How to use these notes

Study one sub-topic at a time. Read the explanations, copy and label diagrams in your notebook, attempt the worked examples without looking at the solution, and complete the practice questions. Practical activities must be carried out under the supervision of a teacher and in accordance with school laboratory safety rules.

- Key terms are explained before they are used.
- Worked examples show the reasoning and the steps, not only the final answer.
- Practical activities include observations, variables and safety reminders.
- At the end of each sub-topic, answer the review questions and check your work using the self-check answers.
- Some scientific models are simplified representations; they help explain evidence but are not exact pictures of microscopic structures.

Annual coverage map

Term	Strand	Sub-strands
1	Mixtures, Elements and Compounds	Structure of the Atom; Metals and Alloys; Water Hardness
2	Living Things and the Environment	Nutrition in Plants; Nutrition in Animals; Reproduction in Plants
3	Living Things and the Environment; Force and Energy	Interdependence of Life; Curved Mirrors; Waves

TERM 1 | STRAND 1: MIXTURES, ELEMENTS AND COMPOUNDS

Chemistry studies the composition of matter, its properties and the changes it undergoes. The particle model helps us explain why substances behave differently.

1. Structure of the Atom

1.1 Matter, elements and atoms

Matter is anything that has mass and occupies space. All ordinary matter is made from particles. An element is a pure substance made of only one type of atom; examples include iron, oxygen, copper and carbon. An atom is the smallest particle of an element that retains the chemical identity of that element.

Atoms are extremely small and cannot be seen using an ordinary school light microscope. Scientists therefore use experiments, measurements and models to explain atomic structure. A model is useful when it explains observations and makes predictions, but it may be revised when new evidence is obtained.

1.2 Subatomic particles

Particle	Relative charge	Relative mass (approx.)	Location	Main role

Proton	+1	1	Nucleus	Determines the atomic number and identity of the element
Neutron	0	1	Nucleus	Adds mass and contributes to nuclear stability
Electron	−1	1/1836	Energy levels/shells around nucleus	Involved in bonding and chemical reactions

The nucleus is the tiny, dense central region of an atom. It contains protons and neutrons and accounts for nearly all the atom's mass. Electrons occupy regions around the nucleus commonly represented at this level as shells or energy levels. Most of the volume of an atom is the space occupied by the electron region.

1.3 Atomic number, mass number and neutral atoms

Atomic number (Z) is the number of protons in the nucleus. It identifies the element. In a neutral atom, the number of electrons equals the number of protons, so the total positive and negative charges balance.

Mass number (A) is the total number of protons and neutrons in the nucleus.

Number of neutrons = mass number − atomic number. For a neutral atom: number of electrons = number of protons.

Worked example: An atom has atomic number 11 and mass number 23. Protons = 11. Because it is neutral, electrons = 11. Neutrons = 23 − 11 = 12.

Worked example: An atom contains 17 protons and 18 neutrons. Its atomic number is 17 and mass number is 17 + 18 = 35. The element with atomic number 17 is chlorine.

1.4 Electron arrangement in shells

For the first twenty elements, a simple school-level shell model commonly places a maximum of 2 electrons in the first shell, 8 in the second and 8 in the third for the basic arrangements studied at this level. Electrons fill the lower-energy inner shell before the next shell is occupied.

Element	Atomic number	Electron arrangement
Hydrogen	1	1
Helium	2	2
Lithium	3	2,1
Carbon	6	2,4
Oxygen	8	2,6
Neon	10	2,8
Sodium	11	2,8,1
Magnesium	12	2,8,2
Chlorine	17	2,8,7

Argon	18	2,8,8
Calcium	20	2,8,8,2

The outermost occupied shell is called the valence shell. The electrons in it are valence electrons. They strongly influence how an atom combines with other atoms. Atoms may gain, lose or share electrons when forming chemical bonds.

1.5 Isotopes and ions

Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons. They have the same atomic number but different mass numbers. For example, carbon-12 and carbon-14 each have six protons, but carbon-12 has six neutrons while carbon-14 has eight.

An ion is an electrically charged atom or group of atoms formed when electrons are transferred. A positive ion (cation) forms when electrons are lost; a negative ion (anion) forms when electrons are gained. The number of protons does not change when an ordinary atom forms an ion.

1.6 Representing an atom

A labelled atomic diagram should show a central nucleus, protons and neutrons in the nucleus, and electrons on shells around it. Use a key or labels so that each particle is distinguishable. Do not draw the nucleus to scale: in reality it is much smaller than the overall atom.

Check your understanding

1. Define an atom and an element.
2. State the charge, approximate relative mass and location of a proton, neutron and electron.
3. An atom has atomic number 13 and mass number 27. Calculate its protons, electrons and neutrons if it is neutral.
4. Write the shell arrangement of oxygen, sodium and chlorine.
5. Explain why isotopes belong to the same element.
6. Distinguish a cation from an anion.

Self-check answers: 1. An atom is the smallest particle retaining an element's identity; an element contains one type of atom. 2. Proton: +1, 1, nucleus; neutron: 0, 1, nucleus; electron: −1, about 1/1836, shells/energy levels. 3. 13 protons, 13 electrons, 14 neutrons. 4. Oxygen 2,6; sodium 2,8,1; chlorine 2,8,7. 5. Same proton number, different neutron number. 6. A cation is positive after electron loss; an anion is negative after electron gain.

2. Metals and Alloys

2.1 Physical properties of metals

Metals are elements that commonly have a shiny surface when freshly cut or polished, conduct heat and electricity, and can be shaped without breaking. Most are solids at room temperature; mercury is a notable liquid exception.

Property	Meaning	Everyday application
Lustre	Shiny appearance	Decorative objects and jewellery
Malleability	Can be hammered or rolled into	Roofing sheets and aluminium

	sheets	foil
Ductility	Can be drawn into wires	Electrical cables
Electrical conductivity	Allows electric current to pass	Copper wiring
Thermal conductivity	Transfers heat readily	Metal cooking pans
Strength	Resists deformation or breaking under load	Bridges and building frames

Not every metal has every property to the same degree. Metals differ in density, hardness, melting point, reactivity and resistance to corrosion. Selection of a metal depends on the job it must perform.

2.2 Chemical behaviour and reactivity

Some metals react readily with oxygen, water or dilute acids, while others react slowly or hardly at all. A more reactive metal tends to lose electrons more readily and forms positive ions. Sodium and potassium are very reactive and must not be handled in ordinary classroom demonstrations. Iron reacts with oxygen and water to form rust; copper develops a green surface layer in moist air over time.

A simplified reaction of a metal with oxygen forms a metal oxide. For example: magnesium + oxygen → magnesium oxide. A simplified reaction of a reactive metal with dilute hydrochloric acid may produce a salt and hydrogen gas. Such reactions should only be demonstrated by a teacher using small quantities and approved procedures.

2.3 Corrosion and prevention

Corrosion is the gradual destruction of a metal through reaction with substances in its environment. Rusting is the corrosion of iron in the presence of both oxygen and water. Salt water generally speeds up rusting.

- Painting or plastic coating keeps air and water away from the metal.
- Oiling or greasing forms a protective barrier, especially on moving parts and tools.
- Galvanising coats iron or steel with zinc, which protects the surface.
- Electroplating deposits a thin layer of another metal, such as chromium or tin.
- Using corrosion-resistant alloys can reduce damage in demanding environments.

2.4 Alloys

An alloy is a material made by combining a metal with one or more other elements. The mixture is designed to improve useful properties such as strength, hardness, resistance to corrosion or ease of casting. Many alloys are mixtures rather than pure compounds with a fixed chemical formula.

Alloy	Main components (simplified)	Useful properties / uses
Brass	Copper + zinc	Workable and corrosion-resistant; fittings and musical instruments
Bronze	Copper + tin	Hard and durable; medals, sculptures and bearings
Steel	Iron + carbon	Stronger and harder than pure iron; buildings, tools and vehicles

Stainless steel	Iron + chromium, often nickel and carbon	Good corrosion resistance; kitchen equipment and medical instruments
Solder	Tin-based alloy, composition varies	Joins metal components in suitable applications

Pure iron is relatively soft compared with many steels. Small amounts of carbon and other elements change how layers of atoms move, giving steel different mechanical properties. The exact properties depend on the alloy's composition and processing.

2.5 Choosing metals and responsible use

When selecting a metal, consider strength, mass, conductivity, corrosion resistance, cost, availability, ease of shaping and environmental impact. Recycling metals saves energy and reduces the need to extract fresh ore. Mining and metal production can disturb land, use large amounts of energy and produce waste, so responsible use and recycling are important.

Practical investigation: comparing metal samples

Aim: Observe and compare selected physical properties of safe, teacher-provided metal samples.

7. Observe the appearance of each sample and record its colour and lustre.
8. Compare the mass of equal-sized samples using a balance if available.
9. With teacher guidance, test whether each sample is attracted to a magnet.
10. Record observations in a table and identify which properties could make each sample useful.

Safety: Do not taste chemicals or samples, do not heat metals without instruction, avoid sharp edges, wash hands after handling and return samples as directed. Do not use unknown metal powders.

Review questions

11. Define malleability and ductility and give one use for each.
12. Explain why copper is used in electrical cables.
13. Describe the conditions needed for iron to rust.
14. Give four methods of preventing corrosion.
15. Define an alloy and explain two reasons for making alloys.
16. Name the main components of brass, bronze and steel.
17. Suggest a suitable material for a cooking pot and justify your choice.

Self-check guide: Malleability means forming sheets; ductility means drawing into wire. Copper is a good electrical conductor and is ductile. Rusting needs oxygen and water. Corrosion barriers include paint, oil/grease, galvanising and electroplating. Alloys are made to improve properties. Brass is copper-zinc; bronze copper-tin; steel iron-carbon. A cooking pot commonly uses a metal with good heat conduction and suitable strength, with a safe handle.

3. Water Hardness

3.1 Meaning and causes

Hard water is water that contains dissolved calcium ions (Ca^{2+}), magnesium ions (Mg^{2+}), or both. These ions commonly enter water when it passes through rocks such as limestone and other mineral deposits. Soft water contains little of these hardness-causing ions.

Hardness is not the same as dirtiness. Water may look clear and still be hard. Hardness alone does not establish whether water is safe to drink; microbiological and chemical safety require separate assessment.

3.2 Temporary and permanent hardness

Temporary hardness is commonly associated with dissolved calcium and magnesium hydrogencarbonates. On heating, these can decompose and form insoluble deposits, so boiling can remove much of this type of hardness.

Permanent hardness is commonly associated with dissolved calcium and magnesium sulphates or chlorides. Ordinary boiling does not remove it effectively. Water softening methods such as ion exchange or suitable chemical treatment may be used.

Type	Common dissolved salts	Effect of boiling
Temporary hardness	Calcium/magnesium hydrogencarbonates	Can be reduced by boiling
Permanent hardness	Calcium/magnesium sulphates or chlorides	Not removed by ordinary boiling

3.3 Effects of hard water

- Soap reacts with calcium and magnesium ions to form an insoluble scum, so more soap is needed to produce lather.
- Scale can form inside kettles, boilers, pipes and heating elements, reducing heat transfer and narrowing water passages.
- Hard water can leave deposits on taps, sinks and utensils.
- Hard water also supplies calcium and magnesium minerals, although the nutritional contribution varies with the water source.

3.4 Testing water hardness

A simple school comparison uses equal volumes of water, equal amounts of soap solution and the same shaking procedure. A sample that forms lather readily with little scum is generally softer than a sample that forms scum and needs more soap. This is a comparative classroom test, not a precise measurement of hardness.

18. Label clean test tubes for the water samples.
19. Measure the same volume of each sample into its tube.
20. Add the same number of drops of soap solution to each.
21. Stopper safely and shake each tube for the same length of time.
22. Compare the amount of lather and scum; record observations.
23. Repeat if necessary to check consistency.

Fair-test variables: keep water volume, soap concentration, number of drops and shaking time constant. Change only the water sample. Do not drink any laboratory sample.

3.5 Softening hard water

- Boiling: reduces temporary hardness by causing some dissolved compounds to form insoluble deposits.

- Ion exchange: a water-softening material exchanges calcium and magnesium ions for other ions, commonly sodium ions, in a controlled treatment system.
- Chemical softening: appropriate chemicals can cause hardness ions to form insoluble precipitates that are then separated. This requires correct dosing and treatment.

Do not assume that boiling makes every hard-water sample soft: permanent hardness remains. Water-treatment systems must be maintained and used according to instructions.

Review questions

24. What is hard water?
25. Distinguish temporary hardness from permanent hardness.
26. Why does hard water waste soap?
27. Explain how scale can affect a kettle.
28. Describe a fair test for comparing two water samples.
29. Which method can reduce temporary hardness? Which method is used for permanent hardness?

Self-check answers: Hard water contains appreciable dissolved calcium and/or magnesium ions. Temporary hardness is commonly due to hydrogencarbonates and can be reduced by boiling; permanent hardness is commonly due to sulphates/chlorides and is not removed by ordinary boiling. Soap scum forms with these ions. Scale reduces heat transfer. A fair test controls volumes, soap amount and shaking time. Boiling reduces temporary hardness; ion exchange or suitable chemical treatment can reduce permanent hardness.

TERM 2 | STRAND 2: LIVING THINGS AND THE ENVIRONMENT

Living organisms carry out life processes, obtain and use materials, reproduce and interact with their surroundings. Nutrition provides materials and energy needed for growth, repair and life activities.

4. Nutrition in Plants

4.1 Photosynthesis

Photosynthesis is the process by which green plants use light energy absorbed by chlorophyll to make glucose from carbon dioxide and water, releasing oxygen. It mainly occurs in chloroplasts in green plant cells, especially in leaves.

Word equation: carbon dioxide + water —(light energy and chlorophyll)→ glucose + oxygen.

Balanced symbol equation: $6CO_2 + 6H_2O$ —(light, chlorophyll)→ $C_6H_{12}O_6 + 6O_2$.

Carbon dioxide enters leaves mainly through stomata. Water is absorbed by roots and transported through xylem. Light is absorbed by chlorophyll in chloroplasts. Oxygen produced can diffuse out through stomata. Glucose may be used in respiration, converted to starch for storage, or used to make cellulose, fats and other substances.

4.2 Leaf structure and adaptations

Leaf feature	How it helps photosynthesis
Broad, flat lamina	Provides a large surface area for light absorption
Thin leaf	Short diffusion distance for gases

Chloroplasts with chlorophyll	Absorb light energy
Stomata	Allow carbon dioxide to enter and oxygen to leave
Veins: xylem	Bring water and mineral salts to the leaf
Veins: phloem	Transport manufactured food to other parts of the plant
Air spaces in spongy mesophyll	Allow gases to diffuse through the leaf

A labelled leaf cross-section should identify upper epidermis, palisade mesophyll, spongy mesophyll and air spaces, lower epidermis, stomata/guard cells and the vascular bundle containing xylem and phloem.

4.3 Factors affecting photosynthesis

- Light intensity: increasing light can increase the rate until another factor becomes limiting.
- Carbon dioxide concentration: increasing carbon dioxide can increase the rate until another factor limits the process.
- Temperature: photosynthesis is controlled by enzymes. A rise in temperature may increase the rate up to an optimum; very high temperatures can damage enzymes and reduce the rate.
- Water availability: shortage of water can cause stomata to close and reduce carbon dioxide entry.

A limiting factor is the factor in shortest supply relative to the organism's needs. Increasing a factor that is not limiting may have little effect on the rate.

4.4 Investigating starch in a leaf

The starch test is used as evidence that photosynthesis has produced stored carbohydrate in a leaf. A teacher should demonstrate or closely supervise the procedure because hot water and ethanol are involved.

30. Destarch a healthy potted plant by keeping it in darkness for an appropriate period under teacher direction.
31. Expose the plant to light for a suitable period.
32. Remove a leaf and boil it briefly in water to kill the cells and stop reactions.
33. Heat the leaf in ethanol in a hot-water bath to remove chlorophyll. Ethanol is flammable and must never be heated directly over a flame.
34. Rinse the leaf in warm water to soften it, then spread it on a white tile.
35. Add iodine solution. A blue-black colour indicates starch; iodine remains yellow-brown where starch is absent.

The destarching step reduces stored starch before the test. A control leaf kept in darkness can help show whether light is needed for starch production.

4.5 Mineral nutrition

Plants also need mineral ions absorbed from the soil solution. Nitrates are needed to make amino acids and proteins; magnesium is needed to make chlorophyll. A shortage of mineral nutrients can cause poor growth or visible deficiency symptoms. Fertilisers supply selected mineral nutrients, but excessive application can pollute water bodies through runoff.

Review questions

36. Write the word equation for photosynthesis.
37. State the roles of chlorophyll, light, carbon dioxide and water.
38. Explain three adaptations of a leaf for photosynthesis.
39. Name four factors that affect the rate of photosynthesis.
40. What does a blue-black colour in the iodine test show?
41. Why must ethanol be heated in a water bath and away from a naked flame?
42. Explain the role of nitrates and magnesium in plant nutrition.

Self-check answers: Photosynthesis uses carbon dioxide and water to make glucose and oxygen. Chlorophyll absorbs light; light supplies energy; carbon dioxide supplies carbon; water supplies hydrogen/electrons and contributes to oxygen release. Leaf adaptations include broad surface, thin structure, chloroplast-rich palisade cells, stomata and veins. Factors: light, carbon dioxide, temperature and water. Blue-black iodine indicates starch. Ethanol is flammable. Nitrates support amino-acid/protein production; magnesium is needed for chlorophyll.

5. Nutrition in Animals

5.1 Nutrients and balanced diet

Animals obtain nutrients by feeding. A balanced diet provides the different nutrients in suitable amounts for an individual's age, activity and state of health.

Nutrient	Main function	Common food sources
Carbohydrates	Main energy source	Cereals, potatoes, rice, maize, fruits
Proteins	Growth and repair; making enzymes and some hormones	Beans, peas, eggs, fish, meat, milk
Fats and oils	Energy store, insulation and cell membranes	Oils, nuts, seeds, dairy products
Vitamins	Regulate body processes and support health	Fruits, vegetables and varied foods
Mineral salts	Build tissues and support body functions	Milk, leafy vegetables, meat, legumes
Water	Solvent, transport medium, temperature regulation	Drinking water and foods
Dietary fibre	Supports movement of food through the gut	Whole grains, vegetables, fruits, legumes

Malnutrition occurs when the diet is inadequate, excessive or unbalanced. It may involve undernutrition, nutrient deficiencies or excess energy intake. A healthy diet should be varied and should take account of local food availability and individual needs.

5.2 Food tests

Food component	Test reagent/procedure	Positive result
Starch	Add iodine solution	Yellow-brown changes to blue-

		black
Reducing sugar	Add Benedict's solution and heat in a hot-water bath	Blue may change through green/yellow/orange to brick-red precipitate, depending on amount
Protein	Biuret reagent (alkali followed by dilute copper sulphate, or prepared Biuret reagent)	Blue changes to lilac/purple
Fat/oil	Ethanol emulsion test: mix sample with ethanol, then add water	Milky-white emulsion

Use small quantities, clean apparatus and separate droppers. Wear eye protection. Do not taste food samples or reagents. Benedict's test requires heating; use a water bath and teacher supervision. Ethanol is flammable.

5.3 The human digestive system

Digestion is the mechanical and chemical breakdown of large, insoluble food molecules into small, soluble molecules that can be absorbed. The alimentary canal is a continuous tube from mouth to anus. Accessory organs such as the liver and pancreas release substances that aid digestion.

Organ	Main function
Mouth	Teeth break food mechanically; saliva moistens food and salivary amylase begins starch digestion
Oesophagus	Moves food to the stomach by peristalsis
Stomach	Churns food; acid provides an acidic condition and helps kill many microbes; protease begins protein digestion
Small intestine	Completes digestion and absorbs soluble nutrients through villi
Liver	Produces bile, which helps emulsify fats
Gall bladder	Stores and releases bile
Pancreas	Produces digestive enzymes and alkaline fluid released into the small intestine
Large intestine	Absorbs much of the remaining water and forms faeces
Rectum and anus	Store and eliminate faeces

Peristalsis is the wave-like contraction and relaxation of muscles in the gut wall that pushes food along. Villi are tiny projections lining the small intestine. They provide a large surface area, have thin walls and contain blood capillaries and a lacteal to aid absorption.

5.4 Digestive enzymes

Enzyme group	Substrate	Products (simplified)
Amylase	Starch	Smaller sugars, ultimately glucose
Protease	Proteins	Peptides and amino acids
Lipase	Fats/lipids	Fatty acids and glycerol

Enzymes are biological catalysts: they speed up reactions without being used up. Each enzyme works best under particular conditions. Very high temperatures can denature an enzyme, changing the shape of its active site. Extreme pH can also reduce enzyme activity.

5.5 Absorption and assimilation

Absorption is the movement of digested soluble nutrients through the wall of the small intestine into blood or lymph. Assimilation is the use of absorbed nutrients by body cells. Glucose can be used in respiration; amino acids are used to build proteins; fatty acids and glycerol can be used to make fats and cell components.

5.6 Food hygiene and digestive health

- Wash hands with soap before preparing or eating food.
- Use clean water and clean utensils; separate raw and cooked foods.
- Cook foods thoroughly and store perishable foods at safe temperatures.
- Eat a varied diet with fibre and drink safe water.
- Seek qualified health advice for persistent digestive symptoms; do not self-diagnose from classroom notes.

Review questions

43. Name the main nutrients and give one function of each.
44. Describe the food tests for starch, reducing sugar, protein and fat.
45. Explain the difference between mechanical and chemical digestion.
46. Describe the roles of the mouth, stomach, small intestine and large intestine.
47. What is peristalsis?
48. Give three adaptations of villi for absorption.
49. Distinguish digestion, absorption and assimilation.

Self-check answers: See the nutrient table and food-test table. Mechanical digestion physically breaks food into smaller pieces; chemical digestion uses enzymes. The mouth chews and starts starch digestion; stomach churns and starts protein digestion; small intestine completes digestion and absorbs nutrients; large intestine absorbs remaining water. Peristalsis is muscular wave movement. Villi have large surface area, thin walls and rich blood supply/lacteals. Digestion breaks down food; absorption moves soluble products into blood/lymph; assimilation is their use by cells.

6. Reproduction in Plants

6.1 Asexual and sexual reproduction

Reproduction is the process by which organisms produce offspring. Asexual reproduction involves one parent and no fusion of gametes; offspring are usually genetically very similar to the parent. Sexual

reproduction involves the fusion of male and female gametes and produces genetic variation among offspring.

Asexual reproduction	Sexual reproduction
Usually one parent	Usually involves male and female gametes
No fertilisation	Fertilisation occurs
Offspring usually genetically similar to parent	Offspring show genetic variation
Can produce many offspring quickly	Often involves pollination and seed formation in flowering plants

6.2 Structure of a flower

Part	Function
Sepals	Protect the developing flower bud
Petals	May attract pollinators through colour, scent and shape
Anther	Produces pollen grains containing male gametes
Filament	Supports the anther
Stigma	Receives pollen
Style	Connects stigma to ovary; pollen tube grows through it
Ovary	Contains ovules
Ovule	Contains the female gamete; develops into a seed after fertilisation

A complete flower commonly has sepals, petals, stamens (male parts) and a carpel/pistil (female part). Some flowers are unisexual and lack either stamens or carpels.

6.3 Pollination

Pollination is the transfer of pollen grains from an anther to a stigma of a flower of the same species. Self-pollination occurs within the same flower or between flowers on the same plant. Cross-pollination is transfer between flowers on different plants of the same species.

Feature	Wind-pollinated flowers	Insect-pollinated flowers
Petals	Often small or dull	Often conspicuous and scented
Pollen	Produced in large quantities; often light and smooth	Often fewer; may be sticky or spiky
Anthers	Often exposed and dangling	Usually positioned to contact visiting insects
Stigma	Often large and feathery	Often sticky and inside the flower
Nectar	Usually absent	Often present

6.4 Fertilisation and seed formation

After compatible pollen lands on a stigma, it may germinate and form a pollen tube. The tube grows down the style towards an ovule. The male gamete travels through the pollen tube and fuses with the female gamete in the ovule. This fusion is fertilisation. After fertilisation, the ovule develops into a seed and the ovary commonly develops into a fruit.

A seed commonly contains an embryo, a food store and a protective seed coat. The embryo develops into the new plant. The food store supports early growth before the seedling can photosynthesise effectively.

6.5 Seed dispersal

Method	Adaptation / example
Wind	Light seeds or seeds with wings/hairs; e.g. dandelion-like seeds
Water	Buoyant fruits or seeds with waterproof coverings; e.g. coconut
Animals	Hooks attach to fur, or fleshy fruits are eaten and seeds later dispersed
Explosive/self-dispersal	Dry fruits split suddenly and scatter seeds

Seed dispersal reduces overcrowding and competition near the parent plant and may allow plants to colonise new areas.

6.6 Germination

Germination is the resumption of growth of an embryo in a seed to form a seedling. Most seeds need water, oxygen and a suitable temperature. Water activates enzymes and softens the seed coat; oxygen is needed for respiration; suitable warmth supports enzyme activity. Light is not essential for germination of many seeds, although it is needed for normal photosynthetic growth after emergence.

In a simple investigation, place equal numbers of similar seeds in different conditions while changing only one factor. For example, compare seeds supplied with water and oxygen at a suitable temperature with seeds deprived of water. Record the number germinating over several days.

Practical: testing conditions for germination

- Use healthy seeds of the same type and similar age.
- Set up labelled containers with equal numbers of seeds.
- Change one condition at a time; keep other conditions as similar as possible.
- Record daily observations, including number germinated and visible changes.
- Calculate percentage germination = (number germinated ÷ total number of seeds) × 100.

Example: 18 out of 20 seeds germinate. Percentage germination = (18 ÷ 20) × 100 = 90%.

Review questions

50. Distinguish sexual from asexual reproduction.
51. Name the male and female reproductive parts of a flower.
52. Define pollination and fertilisation.

53. Give three differences between wind- and insect-pollinated flowers.
54. Explain how an ovule and ovary change after fertilisation.
55. State the three main conditions needed for germination and explain why each is needed.
56. Calculate percentage germination if 24 of 30 seeds germinate.

Self-check answers: Sexual reproduction involves gamete fusion; asexual reproduction does not. Male part is stamen (anther and filament); female part is carpel/pistil (stigma, style, ovary and ovules). Pollination transfers pollen; fertilisation fuses gametes. Wind-pollinated flowers often have exposed anthers, feathery stigmas and abundant light pollen; insect-pollinated flowers often have conspicuous petals, scent and sticky pollen. Ovule becomes seed; ovary commonly becomes fruit. Germination needs water, oxygen and suitable temperature. 24/30 × 100 = 80%.

TERM 3 | STRAND 2: LIVING THINGS AND THE ENVIRONMENT

7. Interdependence of Life

7.1 Ecosystems and habitats

An ecosystem is a community of living organisms interacting with one another and with the non-living parts of their environment. A habitat is the place where an organism lives. A population is all organisms of one species living in a particular area; a community is all the populations of different species living and interacting in that area.

Biotic factors	Abiotic factors
Plants, animals, fungi, bacteria and other living organisms	Light, temperature, water, air, soil, minerals, pH and wind

Organisms have adaptations that help them survive in their habitats. Adaptations may be structural (body features), behavioural (ways of acting) or physiological (internal processes). An adaptation improves survival or reproduction in a particular environment; it is not necessarily useful in every environment.

7.2 Feeding relationships

A food chain shows a feeding pathway through which energy and materials pass. It usually begins with a producer, followed by consumers. An arrow points from the organism being eaten to the organism that eats it, showing the direction of energy transfer.

Example: grass → grasshopper → frog → snake → eagle. Grass is the producer; grasshopper is a primary consumer; frog is a secondary consumer; snake may be a tertiary consumer.

A food web is a network of interconnected food chains. It gives a more realistic picture because most organisms feed on more than one type of organism and may be eaten by several predators.

7.3 Producers, consumers and decomposers

- Producers make organic food, usually by photosynthesis. Green plants and algae are common producers.
- Consumers obtain energy by feeding on other organisms. Herbivores eat plants; carnivores eat animals; omnivores eat both.

- Decomposers, such as many bacteria and fungi, break down dead organisms and waste, returning nutrients to the environment.

7.4 Energy flow and nutrient cycling

Energy enters most ecosystems as sunlight, is captured by producers and passes through feeding relationships. At each transfer, some energy is released to the surroundings as heat through respiration and other life processes. Therefore, less energy is generally available at higher trophic levels.

Energy flows through an ecosystem and is not recycled in the same way as nutrients. Nutrients such as carbon, nitrogen and mineral ions are recycled through organisms, decomposers, soil, water and air.

7.5 Ecological pyramids

A pyramid of numbers represents the number of organisms at each trophic level. A pyramid of biomass represents the total dry mass of organisms at each level. A pyramid of energy represents the energy available at each level over a given period and is upright because energy is lost between trophic levels.

7.6 Relationships between organisms

Relationship	Description	Example
Predation	One organism hunts, kills and eats another	A hawk eating a mouse
Competition	Organisms require the same limited resource	Plants competing for light and water
Mutualism	Both organisms benefit	A pollinating insect obtains nectar while a flower is pollinated
Parasitism	One organism benefits while the host is harmed	A tick feeding on a mammal

7.7 Human activities and ecosystem balance

Human activities can alter habitats and population sizes. Deforestation, pollution, overfishing, poaching, uncontrolled fires, soil erosion and introduction of invasive species may reduce biodiversity and disrupt food webs. Conservation aims to protect species, habitats and ecosystem processes.

- Planting and protecting indigenous trees helps restore habitats and reduce soil erosion.
- Proper waste disposal and treatment reduce pollution of land and water.
- Sustainable farming and fishing reduce pressure on natural resources.
- Wildlife corridors and protected areas help conserve habitats and allow movement of organisms.
- Environmental education encourages responsible decisions by individuals and communities.

Practical field study: a school habitat

57. Select a safe, clearly defined area with teacher approval.
58. Observe and list visible plants, animals and non-living features without damaging the habitat.
59. Use quadrats or transects only when instructed and appropriate.
60. Record observations systematically; avoid touching unknown organisms or disturbing nests.
61. Construct a simple food chain or food web from evidence and explain possible human impacts.

Safety and ethics: do not taste plants, handle unknown animals, collect protected organisms or enter unsafe areas. Wash hands after fieldwork and leave the site undisturbed.

Review questions

62. Define ecosystem, habitat, population and community.
63. Distinguish biotic from abiotic factors and give three examples of each.
64. Construct a food chain with four trophic levels and explain the direction of its arrows.
65. Explain the roles of producers, consumers and decomposers.
66. Why is less energy available at higher trophic levels?
67. Distinguish predation, competition, mutualism and parasitism.
68. Describe four human activities that can disrupt ecosystems and suggest remedies.

Self-check guide: An ecosystem includes organisms and their physical environment; habitat is where an organism lives; population is one species in an area; community is interacting populations. Biotic factors are living; abiotic are non-living. Arrows point in the direction energy moves, from food to feeder. Producers make food; consumers feed; decomposers break down dead matter and wastes. Energy is lost at each transfer, including as heat in respiration. Use the relationship table and conservation actions above.

STRAND 3: FORCE AND ENERGY

8. Curved Mirrors

8.1 Reflection of light

Light travels in straight lines through a uniform transparent medium. Reflection occurs when light strikes a surface and returns into the original medium. The normal is an imaginary line drawn perpendicular to the surface at the point where the ray strikes.

The law of reflection states that the angle of incidence equals the angle of reflection. Both angles are measured from the normal, not from the mirror surface. The incident ray, reflected ray and normal lie in the same plane.

8.2 Types of curved mirrors

Mirror	Shape	Effect on parallel rays	Typical image behaviour
Concave	Reflecting surface curves inward	Parallel rays converge towards a principal focus	Image may be real or virtual; size and orientation depend on object position
Convex	Reflecting surface bulges outward	Parallel rays diverge as if from a virtual focus behind the mirror	Image is virtual, upright and diminished for a real object

Key terms: the principal axis is the line through the centre of curvature and pole; the pole is the midpoint of the mirror surface; the principal focus is where parallel rays meet or appear to come from after

reflection; focal length is the distance from pole to principal focus; centre of curvature is the centre of the sphere of which the mirror is a part.

8.3 Ray diagrams for a concave mirror

Use a ruler and draw principal rays from the top of the object. A ray parallel to the principal axis reflects through the principal focus. A ray passing through the principal focus reflects parallel to the principal axis. A ray directed through the centre of curvature strikes the mirror normally and reflects back along the same path. The point where reflected rays meet locates the image.

Object position (concave mirror)	Image characteristics (qualitative)
Beyond centre of curvature (C)	Between C and F; real, inverted and smaller
At C	At C; real, inverted and same size
Between C and F	Beyond C; real, inverted and larger
At F	Image formed very far away; reflected rays are parallel
Between F and pole (P)	Behind mirror; virtual, upright and larger

A real image is formed where reflected rays actually meet and can be projected onto a screen. A virtual image is formed where rays only appear to meet and cannot be projected onto a screen.

8.4 Uses of curved mirrors

- Convex mirrors are used at road bends and as vehicle side/rear-view mirrors because they provide a wide field of view and upright, diminished images.
- Concave mirrors can produce magnified upright images when an object is close to the mirror, so they are used in some shaving or makeup mirrors.
- Concave reflectors can direct light into a beam when a light source is placed near the focus, as in some torches and headlights.
- Concave mirrors can collect or concentrate light in selected optical devices.

8.5 Practical investigation

Use a ray box, narrow beam, mirror, paper and protractor under teacher supervision. Trace the mirror position, draw a normal at the point of incidence, measure the angles from the normal and compare the angle of incidence with the angle of reflection. Avoid shining bright light into anyone's eyes.

Review questions

69. State the law of reflection.
70. Explain why angles of incidence and reflection are measured from the normal.
71. Distinguish concave and convex mirrors.
72. Describe the image formed by a convex mirror.
73. State the image characteristics when an object is between the focus and pole of a concave mirror.
74. Give two uses of each type of curved mirror and explain the reason for each use.

Self-check answers: Angle of incidence equals angle of reflection. The normal is perpendicular to the surface and is the reference line for the law. Concave mirrors curve inward; convex mirrors bulge outward. A convex mirror gives an upright, virtual and diminished image. An object between F and P in a concave mirror gives a virtual, upright, magnified image behind the mirror. See uses above.

9. Waves

9.1 Meaning and types of waves

A wave is a disturbance that transfers energy from one place to another without the overall transfer of matter from the source to the destination. Particles in a medium may oscillate about fixed positions as the wave passes.

A transverse wave has oscillations perpendicular to the direction of energy transfer. A longitudinal wave has oscillations parallel to the direction of energy transfer. Water surface waves are more complex than the simple transverse model used in introductory diagrams.

Transverse wave	Longitudinal wave
Has crests and troughs	Has compressions and rarefactions
Oscillations are at right angles to energy transfer	Oscillations are parallel to energy transfer
Example: electromagnetic waves; a wave on a taut rope	Example: sound in air; compression wave in a spring

9.2 Wave terms

Term	Meaning	Symbol / unit
Amplitude	Maximum displacement from the rest/equilibrium position	A; metre (m)
Wavelength	Distance between two successive points in phase, such as crest to crest	λ; metre (m)
Frequency	Number of complete oscillations passing a point per second	f; hertz (Hz)
Period	Time taken for one complete oscillation	T; second (s)
Wave speed	Distance travelled by the wave per second	v; metre per second (m/s)

Frequency and period are related by $f = 1/T$ and $T = 1/f$. Wave speed is $v = f\lambda$. These equations apply when the quantities use consistent SI units.

9.3 Worked examples

Example 1: A wave makes 20 complete oscillations in 5 seconds. Frequency = number of oscillations ÷ time = 20 ÷ 5 = 4 Hz. Period = 1 ÷ 4 = 0.25 s.

Example 2: A wave has frequency 5 Hz and wavelength 2 m. Wave speed $v = f\lambda = 5 \times 2 = 10$ m/s.

Example 3: A wave travels at 12 m/s and has frequency 3 Hz. Wavelength $\lambda = v \div f = 12 \div 3 = 4$ m.

9.4 Sound waves

Sound is produced by vibrating sources and travels through a medium as a longitudinal mechanical wave. It cannot travel through a vacuum because there are no particles to transmit the vibrations. In air, sound travels through successive compressions and rarefactions.

Pitch is mainly related to frequency: higher frequency is heard as higher pitch. Loudness is related to the amplitude/intensity of the sound wave; greater amplitude is generally heard as louder sound. Sound level is commonly measured in decibels (dB). Prolonged exposure to loud sound can damage hearing.

9.5 Light and electromagnetic waves

Light is an electromagnetic wave and can travel through a vacuum. Unlike sound, it does not require a material medium. In a vacuum, all electromagnetic waves travel at approximately 3.0×10^8 m/s. Visible light is only a small part of the electromagnetic spectrum.

9.6 Wave behaviour and applications

- Reflection: a wave bounces off a boundary; echoes are reflected sound.
- Refraction: a wave changes direction when its speed changes as it enters a different medium at an angle.
- Diffraction: waves spread out when passing through a gap or around an obstacle; the effect is noticeable when the gap size is comparable to the wavelength.
- Absorption: a material takes in wave energy, often transferring it to internal energy.

Waves are used in communication, medical imaging, musical instruments, navigation and sensing. Different waves have different properties and risks; for example, intense sound can harm hearing and some electromagnetic radiation can damage living tissue.

9.7 Practical investigations

Rope or slinky investigation: generate a transverse pulse on a rope or a longitudinal pulse in a spring. Observe the direction of particle motion relative to the direction the disturbance travels. Do not stretch a spring excessively or aim it at another learner.

Frequency investigation: make a fixed number of oscillations and measure the time with a stopwatch. Calculate frequency using f = number of oscillations ÷ time. Repeat and average results to improve reliability.

Review questions

75. Define a wave.
76. Distinguish transverse and longitudinal waves.
77. Define amplitude, wavelength, frequency and period.
78. A source makes 30 oscillations in 10 seconds. Calculate its frequency and period.
79. A wave has a frequency of 8 Hz and wavelength of 0.5 m. Calculate its speed.
80. Explain why sound cannot travel through a vacuum.
81. Distinguish pitch from loudness.
82. Describe reflection, refraction and diffraction.

Self-check answers: A wave transfers energy without overall transfer of matter. Transverse oscillations are perpendicular; longitudinal oscillations are parallel to energy transfer. Use the wave-terms table. Frequency = 30/10 = 3 Hz; period = 1/3 s ≈ 0.33 s. Speed = 8 × 0.5 = 4 m/s. Sound needs particles in a medium. Pitch relates mainly to frequency; loudness relates to amplitude/intensity. See wave-behaviour definitions.

10. Key Scientific Skills

10.1 Planning a fair investigation

- State a clear aim or question.
- Make a prediction that can be tested.
- Identify the independent variable (what is changed), dependent variable (what is measured) and controlled variables (what is kept the same).
- List apparatus and materials.
- Write a safe, numbered method that another learner could follow.
- Record results with headings, units and appropriate precision.
- Present data in a suitable table or graph.
- Interpret patterns, state a conclusion supported by results and identify limitations or improvements.

10.2 Accuracy, reliability and validity

Accuracy describes how close a measurement is to the true or accepted value. Reliability concerns consistency when a measurement or investigation is repeated. Validity concerns whether the method actually tests the question and controls relevant variables. Repeating measurements and calculating a mean can improve reliability, but does not automatically remove systematic error.

10.3 Laboratory safety

- Follow teacher instructions and read labels before using substances.
- Wear suitable eye protection and protective clothing when required.
- Never taste chemicals or directly smell unknown substances.
- Do not eat or drink in the laboratory.
- Keep flammable liquids such as ethanol away from flames and use a water bath when directed.
- Handle glassware, sharp tools, hot equipment and electrical apparatus carefully.
- Report spills, breakages, injuries and unsafe conditions immediately.
- Dispose of chemicals and biological materials as instructed; wash hands after practical work.

11. Formula and Fact Summary

Topic	Formula / key fact
Atom	Neutrons = mass number − atomic number
Neutral atom	Electrons = protons
Photosynthesis	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (light and chlorophyll)
Percentage germination	(Number germinated ÷ total seeds) × 100%
Frequency	f = number of oscillations ÷ time
Period	T = 1/f
Wave speed	$v = f\lambda$
Law of reflection	Angle of incidence = angle of reflection

12. End-of-Year Revision Questions

83. An atom has atomic number 16 and mass number 32. State the number of protons, neutrons and electrons in a neutral atom.
84. Explain the difference between an isotope and an ion.
85. Give three physical properties of metals and link each to an application.
86. Explain why steel is an alloy and state one advantage of making it.
87. Compare temporary and permanent hardness and describe a method for reducing each.
88. Write the photosynthesis word equation and explain the role of stomata.
89. Describe the starch test in a leaf and state the positive result.
90. Explain the roles of amylase, protease and lipase in digestion.
91. Describe how the structure of a villus supports absorption.
92. Explain the sequence from pollination to seed formation.
93. State the conditions needed for germination and explain their roles.
94. Construct a food chain and identify the producer and two consumer levels.
95. Explain why energy decreases along a food chain.
96. State the law of reflection and describe the image formed by a convex mirror.
97. A wave completes 45 oscillations in 15 seconds. Calculate its frequency and period.
98. A wave has frequency 12 Hz and wavelength 0.25 m. Calculate its speed.

Suggested answers

99. 16 protons, 16 electrons and 16 neutrons.
100. Isotopes have the same proton number but different neutron numbers; ions are charged because electrons have been gained or lost.
101. Examples: conductivity for wiring; malleability for sheets; ductility for wires; strength for structures.
102. Steel is mainly iron with carbon and may contain other elements; alloying improves properties such as strength or hardness.
103. Temporary hardness is commonly caused by hydrogencarbonates and can be reduced by boiling; permanent hardness is commonly caused by sulphates/chlorides and needs ion exchange or suitable chemical treatment.
104. Carbon dioxide + water → glucose + oxygen; stomata allow gas exchange.
105. Boil in water, decolourise in ethanol using a hot-water bath, rinse and add iodine; blue-black indicates starch.
106. Amylase breaks down starch; protease breaks down proteins; lipase breaks down fats.
107. Villi provide large surface area, thin walls and blood capillaries/lacteals for transport.
108. Pollen lands on stigma, pollen tube grows down style, male gamete fuses with female gamete; ovule becomes seed and ovary commonly becomes fruit.
109. Water, oxygen and suitable temperature; water activates processes, oxygen supports respiration and warmth supports enzyme activity.
110. Example: grass → grasshopper → frog; grass is producer, grasshopper primary consumer, frog secondary consumer.
111. Energy is transferred to the surroundings at each trophic level, including during respiration.
112. Angle of incidence equals angle of reflection; convex mirror gives a virtual, upright, diminished image.

113. Frequency = 45/15 = 3 Hz; period = 1/3 s ≈ 0.33 s.
114. Speed = fλ = 12 × 0.25 = 3 m/s.

Curriculum and teacher-use note

This is an original teacher-prepared learning resource organised around the Grade 9 Integrated Science annual coverage map used for the accompanying school scheme of work: Structure of the Atom; Metals and Alloys; Water Hardness; Nutrition in Plants; Nutrition in Animals; Reproduction in Plants; Interdependence of Life; Curved Mirrors; and Waves. It is intended to support, not replace, the current KICD Grade 9 Integrated Science Curriculum Design, approved learner materials, teacher judgement and school safety procedures. Teachers should verify depth, terminology, lesson sequence and practical requirements against the current official design and adapt examples to local context.

Learners should use these notes alongside classroom demonstrations, supervised investigations, approved textbooks and teacher feedback. Never perform chemical, heating, electrical or field activities without appropriate supervision.